

PROFESSIONAL SUMMARY

Senior Data Engineer with 9 years of experience building data pipelines, data models, and cloud data platforms across retail, healthcare, and financial services. Experienced in AWS and Google Cloud, with strong hands-on expertise in Snowflake, BigQuery, SQL, Python, Apache Spark, and Airflow. Delivered large-scale data engineering solutions supporting \$250M+ in daily claims processing and \$1.2B in annual subscription revenue, with additional experience in cloud migration, Snowflake Data Clean Rooms, data governance, DevOps, and automation.

TECHNICAL SKILLS

- Cloud Platforms:** AWS, Google Cloud
- Data Warehousing:** Snowflake, BigQuery, Hadoop, Teradata
- Programming & Scripting:** Python, PySpark, SQL, Shell Scripting
- Orchestration:** Apache Airflow, Control-M, NiFi
- Data Engineering Frameworks:** Apache Spark, Hive, Hbase, kafka
- BI Tools:** Looker Studio Pro, Power BI
- CI/CD:** Git, GitHub, Bitbucket, Terraform, Linux/Unix
- Architecture & Governance:** Data Modeling, Medallion Architecture, Data Lineage, Data Quality

PROFESSIONAL EXPERIENCE

- Best Buy India**

Software Engineer II – Data

Bangalore, India

Apr 2025 – Present

 - Designed and built a BigQuery data model to consolidate active subscription customers for Best Buy, enabling tracking of \$1.2B in annual subscription revenue and providing a unified view of customer subscription data for reporting and analysis.
 - Architected and successfully developed a complex BigQuery data model for Best Buy’s conditional offer sales, integrating 6 product-category-specific tables with intricate business logic to calculate offer amounts across products with varying warranty structures and frequently changing business rules.
 - Designing and implementing a Snowflake Data Clean Room for Best Buy to establish a secure environment for joint analytics with vendor partners without exposing underlying raw data.
 - Successfully modernized and rebuilt data pipelines in Google Cloud Platform (GCP) to migrate on-premises data assets to the cloud, reducing infrastructure dependency and lowering operational costs by approximately 30%.
 - Engineered and deployed a Google Cloud Workflow to process 5 customer subscription assets in parallel, serving as the source of truth for more than 20 years of customer subscription lifecycle data.
 - Successfully implemented DevOps and cloud foundation capabilities as part of the Cloud Data Foundation team, including secure project provisioning, secret management, Workload Identity Federation (WIF), and OAuth integration.
- Carelon Global Solutions**

Senior Software Engineer – Data Engineering

Bangalore, India

Jan 2021 – Mar 2025

 - Designed and successfully developed an ETL pipeline using Python and SQL on Google Cloud to monitor and process \$250M of finalized claims daily for the MBM Commercial accounting model.
 - Architected and built an AWS Glue-based ETL framework to support incremental and bulk data movement across Amazon S3, enabling secure and efficient exchange of 200 TB of incentives data with external vendors.
 - Strategically designed and reviewed enterprise data models and architecture solutions, incorporating data governance, security, and PHI/PII risk mitigation requirements for US-based clients.
 - Successfully optimized complex Snowflake SQL workloads on large datasets and contributed to code reviews, data quality improvements, and secure development practices.
 - Successfully managed and mentored a team of 8+ associates on Big Data technologies, Hadoop, data lake architecture, and development practices, improving technical delivery across the team.

- Successfully enhanced a Hive partition cleanup framework, reducing execution time from approximately 30 minutes to less than 1 minute per table (approximately 98% reduction) and earning the "Best of the Month" award at Accenture BDF.
- Engineered and applied a PySpark-based test automation framework using Python and shell scripting for validating the conversion of StreamSets ETL pipelines to Spark jobs, reducing testing effort from 8 hours to under 30 minutes per table.
- Redesigned and successfully enhanced a centralized RDBMS-to-Hadoop ingestion framework using Shell, PySpark, Sqoop, Hive, and HBase, scaling it to support 1,800+ production data pipelines.
- Successfully engineered an automated auditing framework to validate HBase entries across 1,600+ StreamSets pipelines, with automated failure logging and email notifications that eliminated nearly 2 days of manual effort.
- Designed and built an automation tool to scan audit data across HBase and identify missing data loads between Hadoop and Snowflake, supporting validation across 1,000+ tables.

DATA ENGINEERING CASE STUDY

Clickstream Funnel Analysis – System Design

Documentation

- Designed a GCP-based clickstream funnel architecture capable of processing 100M+ daily behavioral events, using Pub/Sub, Cloud Storage, BigQuery, and Cloud Composer within a decoupled medallion/lakehouse architecture.
- Designed hot and cold ingestion paths using BigQuery and Parquet-based Cloud Storage to balance low-latency analytics, replayability, and storage efficiency.
- Designed the data-processing approach for anonymous-to-authenticated identity stitching and chronological funnel sequencing, enabling end-to-end user journey and conversion analysis.

EDUCATION

B.E. – Electrical & Electronics Engineering
Panimalar Engineering College, Anna University, Chennai

Aug 2013 – Jun 2017
First Class

CERTIFICATIONS

AWS Certified Data Engineer – Associate	<i>Dec 2024 – Dec 2027</i>
AWS Certified AI Practitioner	<i>Nov 2025 – Nov 2028</i>
Google Cloud Professional Data Engineer	<i>Feb 2026 – Feb 2028</i>
Google Cloud AI Leader	<i>Jan 2026 – Jan 2029</i>

ACHIEVEMENTS & LEADERSHIP

- Toastmasters – Carelon Orators:** Vice President, Membership (Jul 2023 – Dec 2023)
- Toastmasters – Carelon Orators:** Sergeant at Arms (Jul 2022 – Jun 2023)
- HackerRank:** 5-Star Rating in SQL and Python